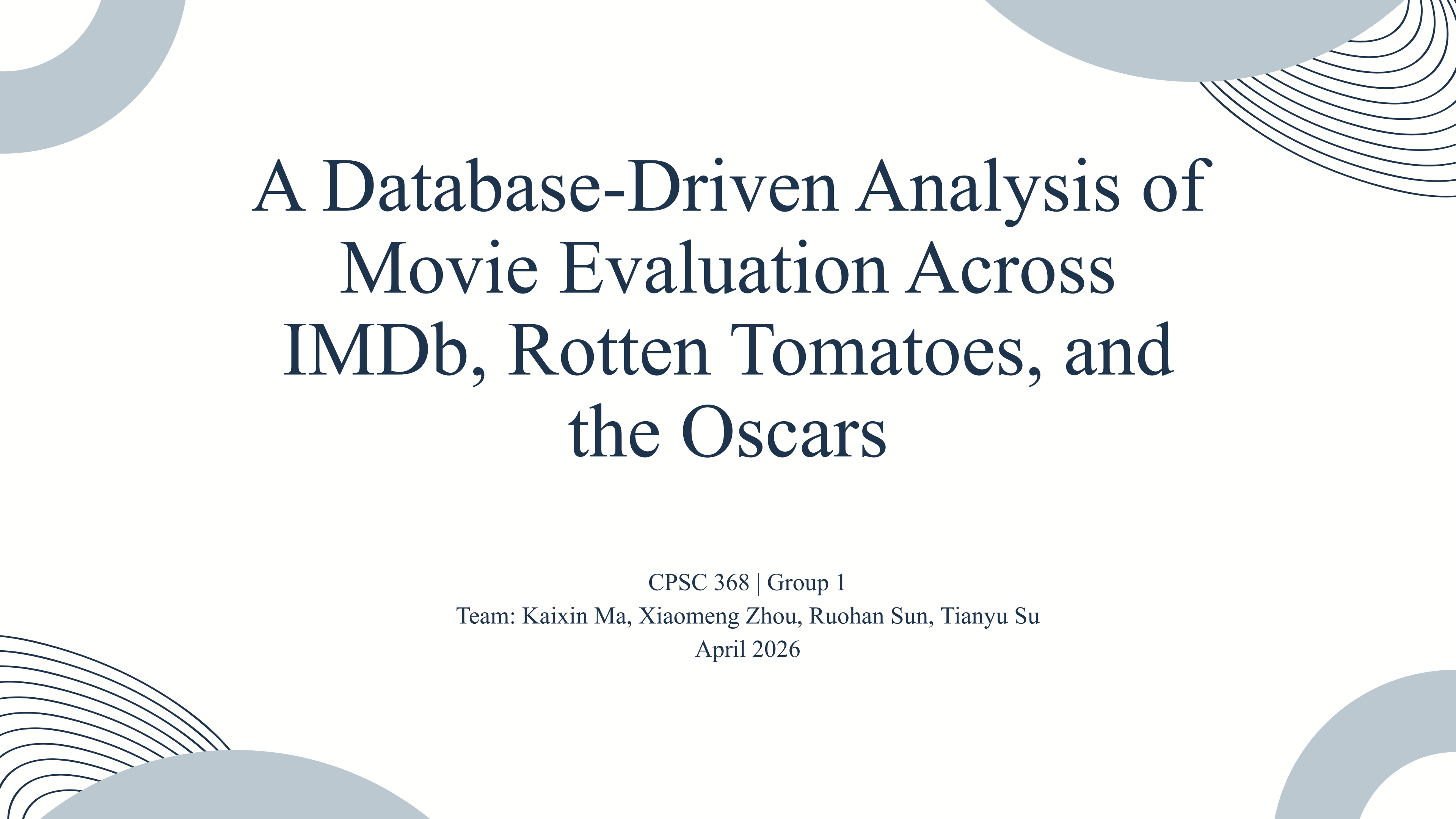




A Database-Driven Analysis of Movie Evaluation Across IMDb, Rotten Tomatoes, and the Oscars

CPSC 368 | Group 1

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Introduction

- There are multiple movie evaluation platforms, such as:
 - IMDb (audience ratings)
 - Rotten Tomatoes (audience & critics)
 - Oscars (award recognition)
- Each platform uses different standards and evaluation criteria
 - Audience vs critics vs industry awards
 - Different scoring systems and perspectives
- As a result:
 - Movie ratings are not directly comparable
 - It is unclear what a rating actually reflects:
 - Quality?
 - Popularity?
 - Critical recognition?

This creates confusion for both audiences and industry.

Motivation

- Movie ratings strongly influence:
 - Audience decisions (what to watch)
 - Film reputation
 - Financial success
- However:
 - Different platforms may tell different stories
 - A high rating on one platform $\neq$ high rating on another
- Key problem:
 - Lack of integrated understanding across platforms
- We want to combine multiple sources to understand:
 - How ratings relate to each other
 - What they actually represent

Research Questions

We study movies from 2016–2025 and focus on three key relationships:

- Platform Consistency:
 - Are IMDb ratings consistent with Rotten Tomatoes audience scores?
- Awards vs Ratings
 - Does Oscar recognition predict higher IMDb ratings (controlling for popularity & genre)?
- Popularity vs CriticsIs
 - movie popularity (votes) related to critic scores?

Overall question:

Do different evaluation systems tell the same story about a movie?

Data Sources

- **IMDb Dataset**
 - Audience ratings (0–10), number of votes
 - Widely used and authoritative movie database
- **Rotten Tomatoes Dataset** (Kaggle, from official API)
 - Audience & critic scores
 - Provides both public and professional evaluations
- **Oscar Dataset** (Kaggle, based on official records)
 - Award and nomination information
 - Represents industry recognition

These datasets are complementary:

audience + critics + awards → full picture of movie evaluation

Data Cleaning

For all datasets, we remove missing values, filter 2016–2025.

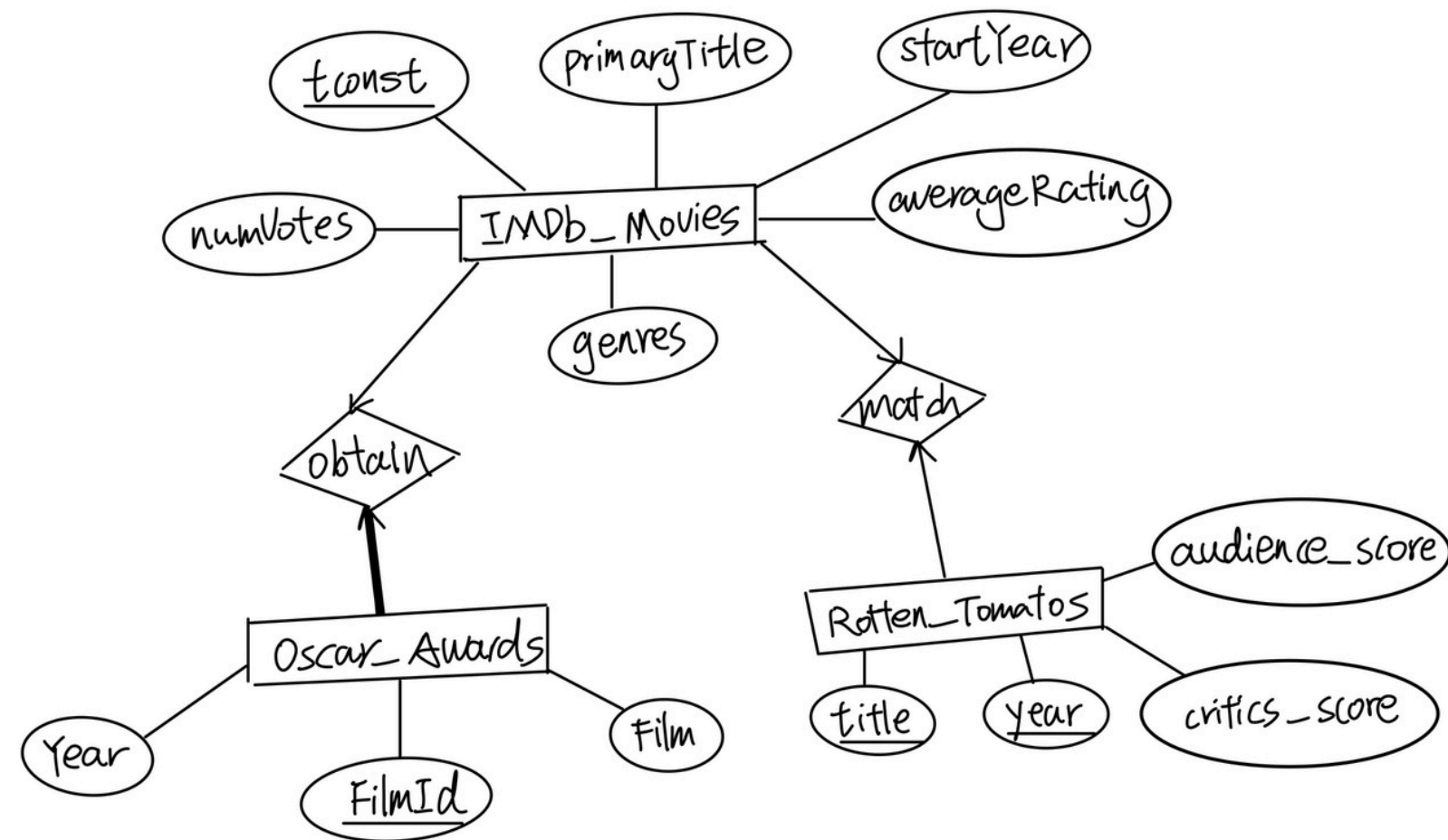
- **IMDb Dataset**
 - Keep movies with ≥ 2000 votes (ensure reliability)
- **Rotten Tomatoes Dataset**
 - Convert JSON to table format
 - Rescale scores to match IMDb (0–10)
- **Oscar Dataset**
 - Remove unmatched records (missing IDs)
 - Deduplicate films
- **Integration**
 - Standardize titles (lowercase, remove punctuation)
 - Match movies by title & year
 - Merge into a unified dataset for analysis

Methodology

- Workflow steps:
 - Integrated three datasets: IMDb, Rotten Tomatoes, and Oscars for movies released from 2016 to 2025
 - Cleaned and standardized titles, years, and score formats to support cross-dataset matching
 - Queried both Oracle and MongoDB using Python, with filtering, joins, and aggregation tailored to each research question
 - Used descriptive and exploratory analysis to examine relationships among ratings, popularity, critic scores, and Oscar recognition
 - Visualized results with boxplots, scatter plots, hexbin plots, and grouped line charts for interpretation

Database Design

- 3 relational tables
- IMDb as core table
- Primary key: tconst
- Oscars linked by FilmId
- RT matched by title + year
- Supports joins and analysis



Database Design

MongoDB (Document-Oriented Database):

- In MongoDB, each movie was stored as a single document.
- IMDb, Rotten Tomatoes, and Oscars data were combined into one record.
- We chose embedded data instead of references because the project was centered on the movie as a single entity.
- This created a self-contained and movie-centered structure.
- The design was intuitive and convenient for retrieving information about one movie.

Database Comparison

Final Decision

- We chose Oracle as the final database. Our project integrates multiple datasets (IMDb, Rotten Tomatoes, Oscars).
- Requires: joins, grouping, comparisons.
- Oracle is better for structured analytical queries.
- Our project focuses on analysis across datasets.

Database Comparison

Oracle vs MongoDB

Oracle:

- Structured relational model
- Easy joins and group queries
- Better for cross-dataset analysis

MongoDB

- Document-based, flexible
- No joins needed
- Better for simple, single-movie queries

MongoDB is convenient for simple queries, but Oracle performs better for complex analysis.

Database Comparison

Reflection

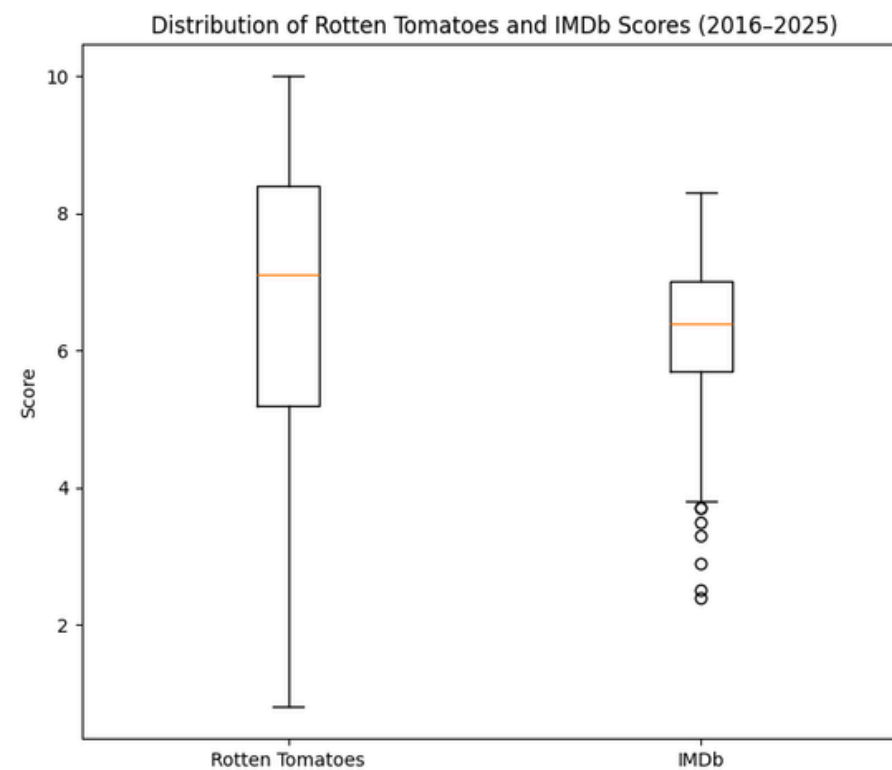
- Q1 → MongoDB (stored directly within the same document)
- Q2 → Oracle (requires more structured comparisons centered on Oscar recognition, popularity, and genre.)
- Q3 → Both workable

Key Insight:

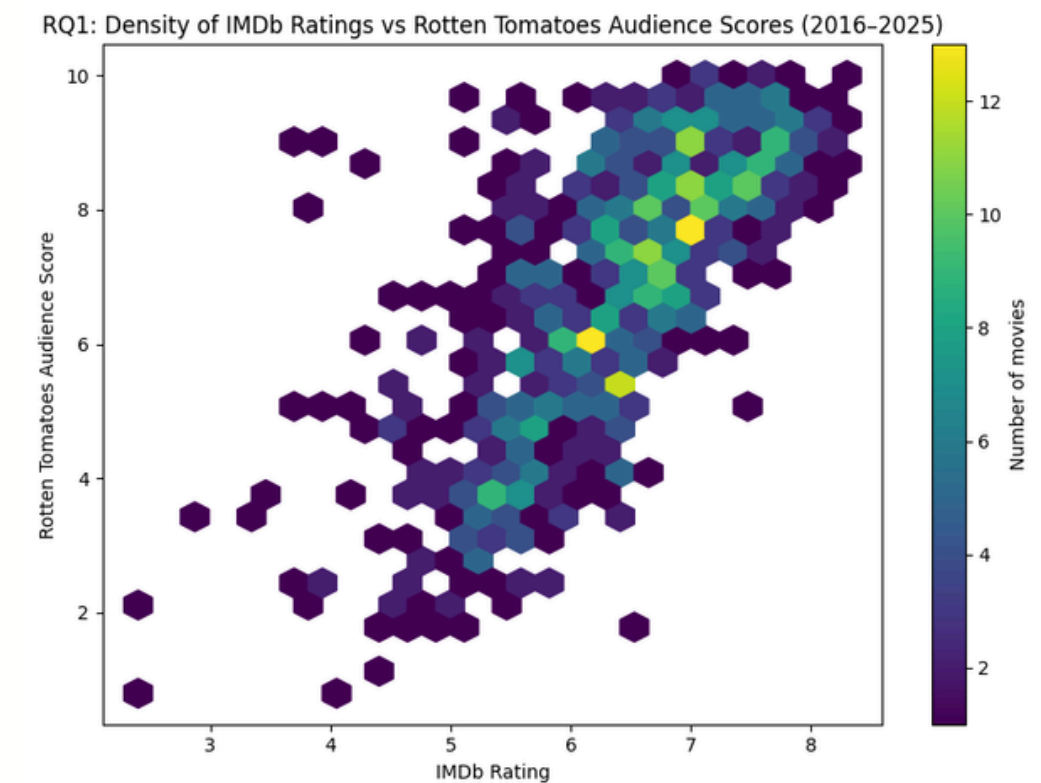
- MongoDB → flexible data
- Oracle → structured comparison

Final choice depends on: data structure and query complexity. The database choice depends on the type of data and queries, but for our project, Oracle is the better fit.

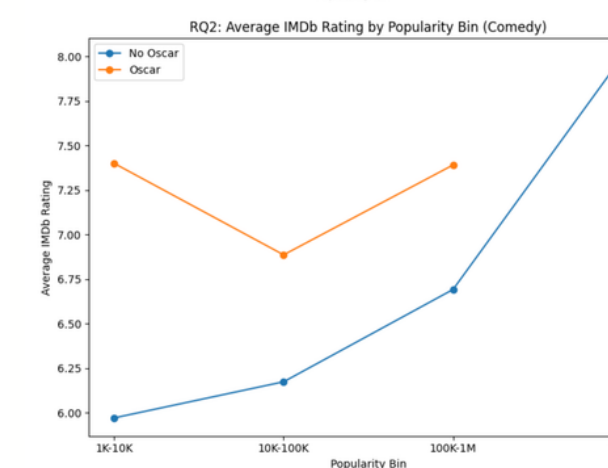
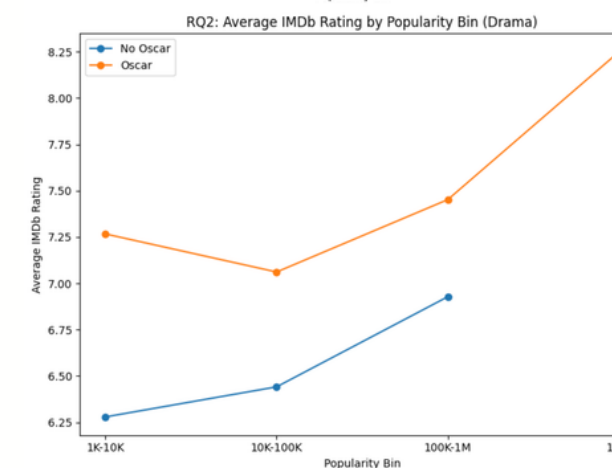
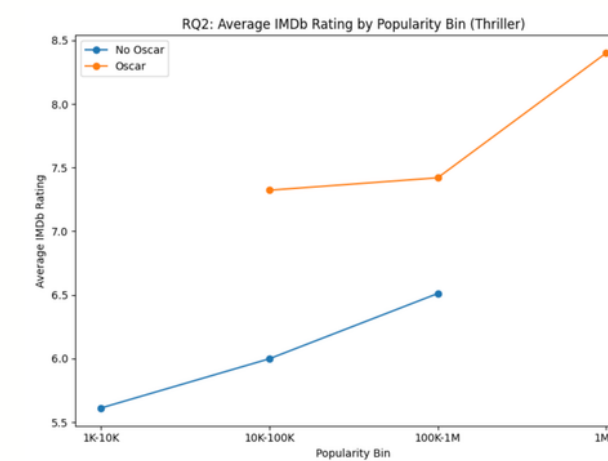
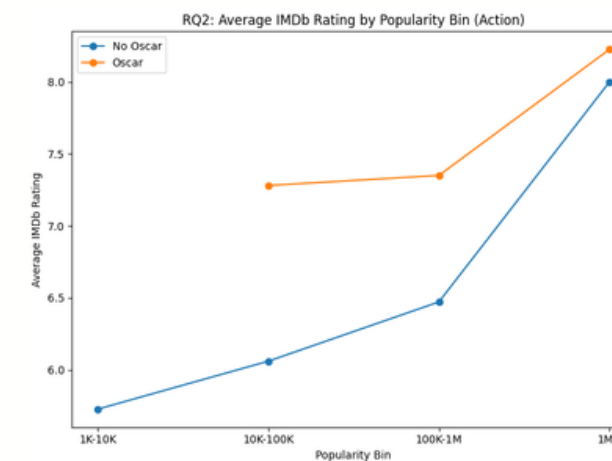
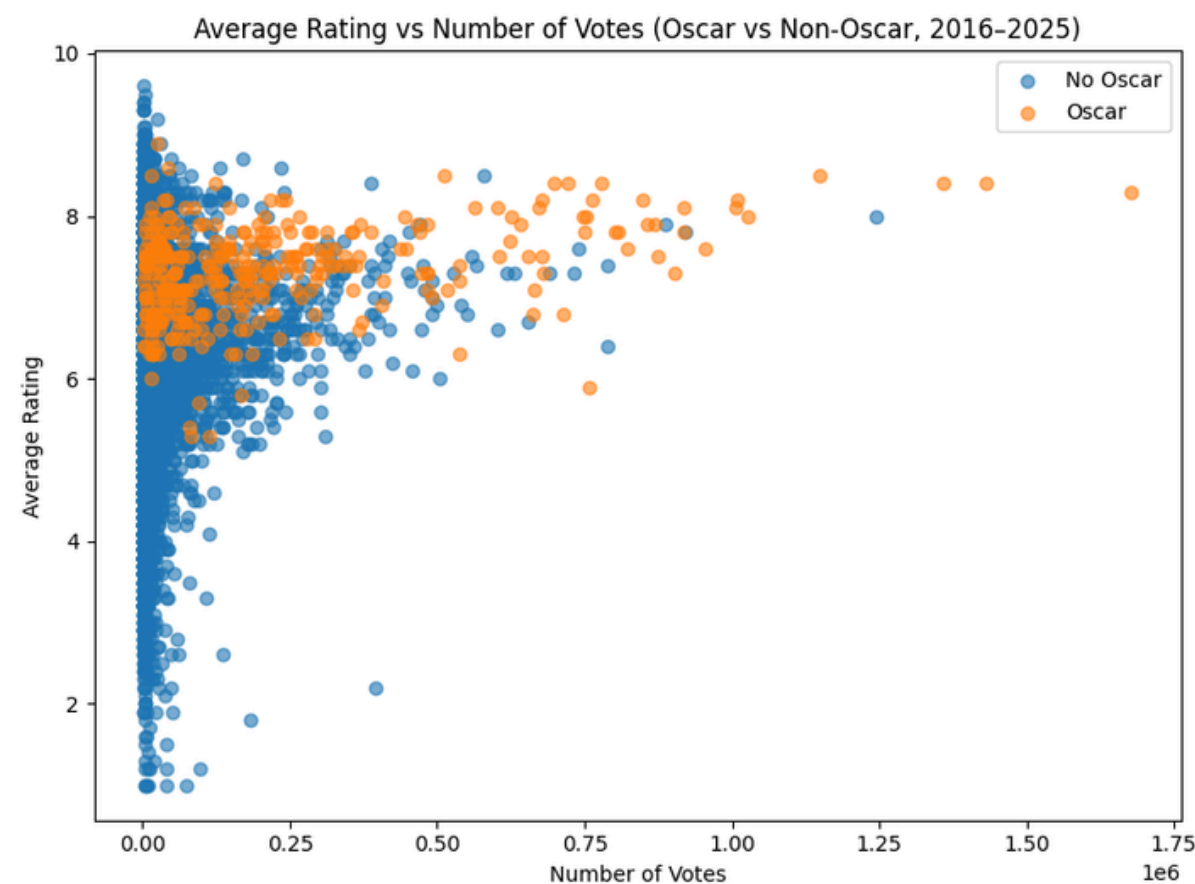
Results — RQ1: IMDb vs Rotten Tomatoes Audience Scores



- Positive correlation between IMDb and Rotten Tomatoes audience ratings
- IMDb ratings concentrated mostly between 5.5 and 7
- Rotten Tomatoes scores more widely spread across the scale
- Some discrepancies highlight rating differences by platform

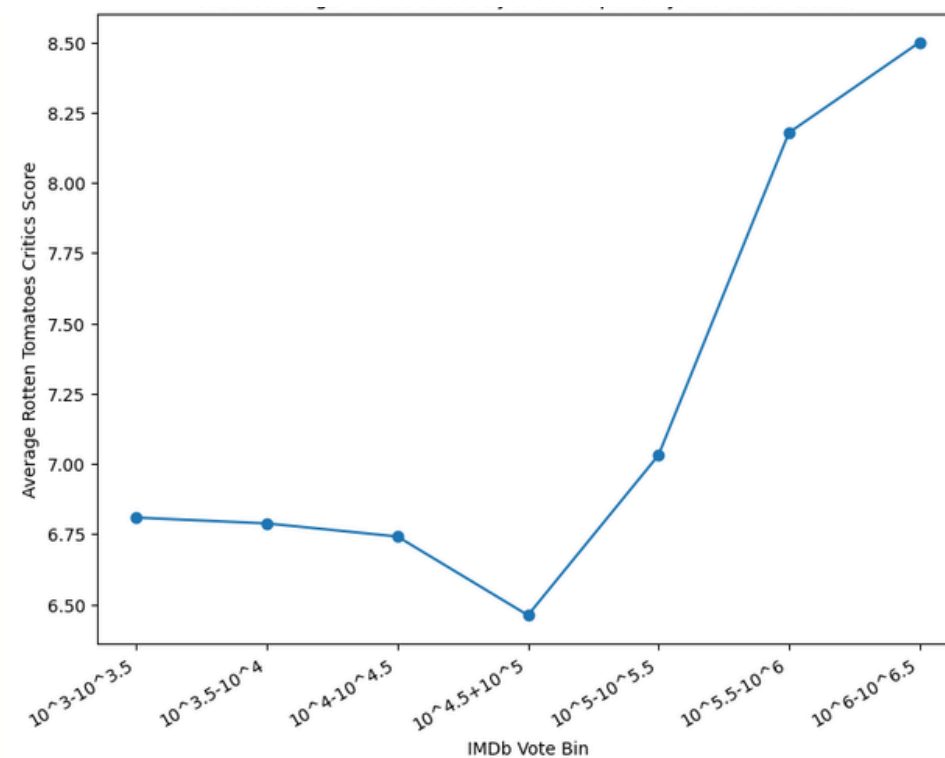
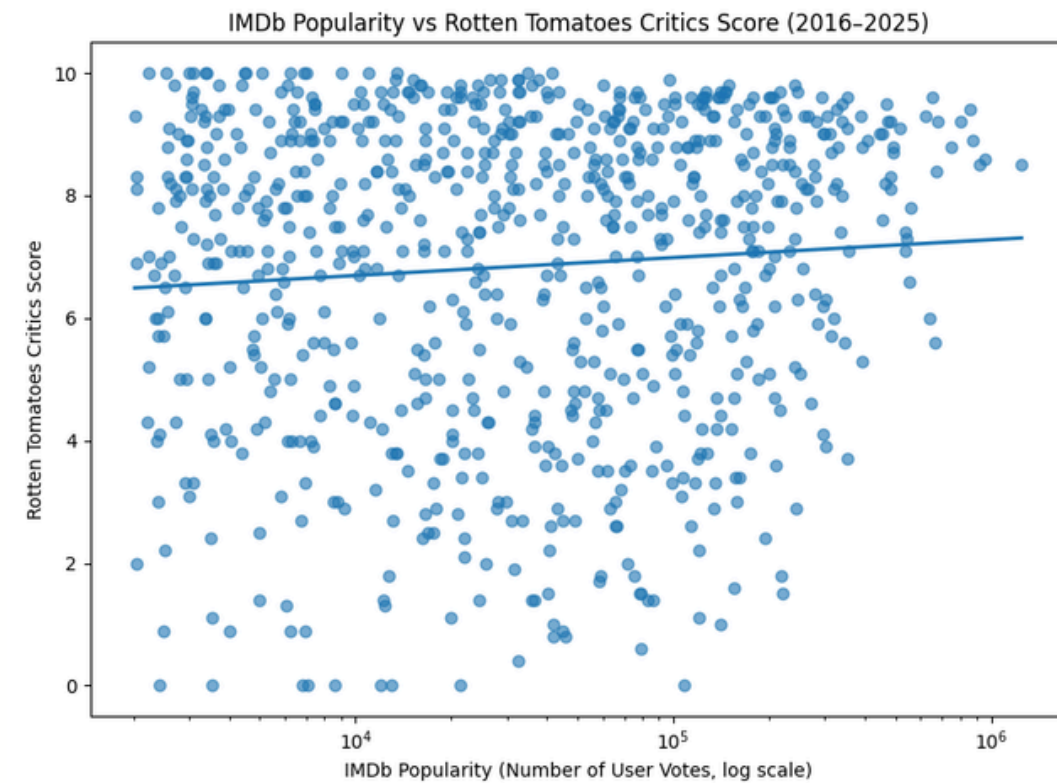


Results — RQ2: Oscar Recognition Predicting IMDb Ratings



- Oscar-winning movies tend to have significantly higher IMDb ratings
- This association remains after controlling for genre and popularity

Results — RQ3: IMDb Popularity vs Critic Scores



- Weak positive correlation observed between IMDb popularity and Rotten Tomatoes critic scores
- Popularity is not a strong predictor of critic ratings
- Higher vote counts do not guarantee favorable critical reception
- Placeholder: Insert scatter plot showing trend here

Limitations

- Data integration issues: possible mismatches across datasets.
- Data filtering: removed missing values & low-vote movies (less than 2000 votes), may introduce selection bias.
- Not total participation constraint: some movies are not present in the IMDb datasets, but are present in Rotten Tomatoes and the Oscars.
- Only descriptive analysis: cannot infer causality.

Future improvements

- Improve cross-dataset matching.
- Add more variables: film budgets, box office performance, distribution bution platforms.
- Use more advanced models: control for confounding factors.
- Future research could explore whether these correlations differ across film genres or whether the link between critics' acclaim and awards is stronger than that based on audience ratings.



Thank You!

Questions? We'd love to hear from you.